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Do private schools increase academic achievement? Evidence from France

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ABSTRACT

This article investigates the effect of private lower secondary schools on student achievement in France. I use propensity score matching on a large French database to estimate the effect of enrollment in a private school on academic achievement as measured by ninth-grade test scores in three school subjects. I find that private school attendance has a large and significant effect on educational success. Boys' (girls') scores in private school were between 0.193 (0.138) and 0.222 (0.198) standard deviations higher on standardized tests in ninth grade. A series of checks confirm the robustness of these results.

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1. Introduction

For several decades, there has been a growing scholarly interest in the comparative effectiveness of private and public education systems for student achievement. While contrasting results have been found – depending on the country studied, the methodology used, the educational level, or the outcome of interest (see, for example, Chabrier, Cohodes, and Oreopoulos 2016) – there is a relatively broad consensus in the literature that private schools outperform public schools (Newhouse and Beegle 2006).

In the USA, the report of Coleman, Hoffer, and Kilgore (1982) is the starting point of an extensive literature. These authors showed that students in private schools have higher academic achievement and that private schools contribute to desegregation. Nevertheless, early studies all confronted with the chief empirical difficulty facing researchers trying to estimate the effect of the private sector on educational success: the existence of a potential selection bias. The question of the capacity of a method to take into account the non-random selection of the students who enroll in private schools is thus central. Samples of private school students differ fundamentally from those of public school students. Obviously, in the USA and elsewhere, enrollment in private school may depend on the ability to pay large fees, physical location in a higher average SES (socioeconomic status) location, etc. In particular, this implies that, on average, private students come from more privileged social backgrounds, their parents are generally better educated, and they often have better academic skills (Thapa 2015). Thus, the effect of private schooling cannot be assessed by simply comparing the outcomes of students enrolled in private and public schools.

To overcome this issue, different methods have been used to estimate the effect of private schools on student achievement. To account for the selection bias associated with being enrolled in a Catholic school, Evans and Schwab (1995) instrument the choice of attending a Catholic

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school using (i) Catholic religion, (ii) school attendance in a predominantly Catholic area, and (iii) a set of variables that the authors constructed by interacting Catholic religion with religious attendance variables as instruments. Based also on the fact that the decision to attend Catholic schools is not exogenous, Neal (1997) considers that using religious affiliation as an instrument for Catholic school attendance may be problematic. To take into account the selection bias linked to Catholic school attendance, he use (i) Catholic adherents as a fraction of the local population and (ii) Catholic schools per square mile as instruments. However, Vandenberghe and Robin (2004) underline that the main obstacle to the use of instrumental variables estimation to measure the effects of private schooling is the need to find a valid instrument. The natural method for dealing with selection bias would be to randomly assign students to private and public schools. It is generally impossible to conduct such an experiment. But where there is more demand than supply for charter school places, schools may use a lottery to determine who will be offered a place, a fact that has been exploited in many studies, especially in the USA (Chabrier, Cohodes, and Oreopoulos 2016). This type of natural experiment approximates the conditions of a randomized trial in some respects. However, as stressed by Spees and Lauen (2019), the validity of this type of study is limited by the fact that the results can only be applied to charter schools with more applicants than available places. Moreover, this method can only be implemented in countries where such lottery systems are used, such as the USA, Colombia (Angrist et al. 2002), and India (Muralidharan and Sundararaman 2015). Another common way of estimating the effect of charter schools is to use fixed-effects modeling with longitudinal data (Epplé, Romano, and Zimmer 2016). These models compare students' outcomes before and after enrollment in a charter school. The advantage of this method is that it does not depend on the existence of a lottery that assigns students to charter schools. But the fixed-effects approach provides estimates only for students moving from public to charter schools ('switchers'), limiting the generalization of the results. This method does not provide an estimate for students who attend only charter schools. Moreover, switchers and non-switchers differ: students who switch schools are a small and unrepresentative population (Spees and Lauen 2019). In addition, this approach does not address potentially time varying unobservables (such as an increase in family income that could lead to additional educational resources and increase the likelihood of attending private school).

An alternative method that relies on selection on observables¹ reducing the potential role of unobservables has emerged in the literature (Epplé, Romano, and Zimmer 2016; Spees and Lauen 2019): propensity score matching (PSM) (see, for example, Anand, Mizala, and Repetto 2009; Azam, Kingdon, and Wu 2016; Chudgar and Quin 2012; Nguyen, Taylor, and Bradley 2006; Sass et al. 2016; Thapa 2015; Wamalwa and Burns 2018; Vandenberghe and Robin 2004). This method is indicated when students are not randomly assigned to private schools. This technique is also particularly more suitable when there is a large number of covariates (Caliendo and Kopeinig 2008) to control for observed variables that affect both the probability of enrolling in private schools and the outcomes. As stressed by Sianesi (2004), this approach requires a detailed understanding of the institutional configuration from selection to treatment. Nevertheless, the results obtained by PSM are not informative if something unobservable is relevant to both enrollment in private school and outcomes. To assess the potential effect of an unobserved component on the estimation results obtained by PSM, it is possible to implement robustness checks (see Section 4.3).

Regardless of the method, various studies have highlighted the positive impact of attending charter school in the USA on test scores (Abdulkadiroğlu et al. 2011; Angrist et al. 2010, 2016; Angrist, Pathak, and Walters 2013). Some studies have found that the nature of the advantage of attending charter or private school may depend on grade level and educational subject (Dobbie and Fryer 2011, 2013; Rouse 1998; Sander 1996; Zimmer et al. 2012). While the existing literature mainly consists of research in the USA, there are studies highlighting the private school advantage in test scores in Canada (Lefebvre, Merrigan, and Verstraete 2011), Chile (Anand, Mizala, and Repetto 2009), Colombia (Angrist et al. 2002; Cox and Jimenez 1991), India (Azam, Kingdon, and Wu 2016; Kingdon 1996; Thapa 2015), Kenya (Wamalwa and Burns 2018), and Tanzania (Cox and Jimenez

1991). Moreover, some studies have found no advantage, or even a negative advantage, of attending charter or private school on achievement (Abdulkadiroğlu, Pathak, and Walters 2018; Bifulco and Ladd 2006; Bettinger 2005; Chudgar and Quin 2012; Newhouse and Beegle 2006; Ni and Rorrer 2012). Beyond the question of test scores, the literature highlights the positive advantage of charter or private schooling on rates of graduation from high school and/or university attendance (Altonji, Elder, and Taber 2005; Angrist et al. 2016; Booker et al. 2011; Davis and Heller 2019; Evans and Schwab 1995; Figlio and Stone 1999; Nguyen, Taylor, and Bradley 2006; Sass et al. 2016), dropout rates (Sander and Krautmann 1995), and discipline (Cullen, Jacob, and Levitt 2006; Imberman 2011). Furthermore, the literature shows that the advantages of enrollment in charter or private schools can be mixed depending on the level of school (Baude et al. 2020; Buddin and Zimmer 2005; Hanushek et al. 2007; Sass 2006), the number of years of enrollment in school (Booker et al. 2007), the size of the urban area (Angrist, Pathak, and Walters 2013; Neal 1997; Zimmer and Buddin 2006), the student's grade level when they register in school (Hoxby and Rockoff 2004), the US state where the student is enrolled (Zimmer et al. 2012), or the country (Vandenbergh and Robin 2004). Finally, a number of studies have looked at the advantages of charter or private schooling on non-educational outcomes such as earnings (Binelli and Rubio-Codina 2013; Sass et al. 2016), marital status (Angrist et al. 2002), propensity to commit a crime and to participate in elections (McEachin et al. 2020), pregnancy and imprisonment (Dobbie and Fryer 2015), wellbeing (Green et al. 2014) and likelihood that teenagers use or sell drugs, commit property crime, have sex, join gangs, attempt suicide, or run away from home (Mocan and Tekin 2006).

While there is a considerable international literature on the effects of private schooling on student achievement, there has been little research on this subject in France. This question is all the more distinct given that private schools in France are very different from private schools in other countries. The majority of private lower secondary schools in France are under contract with the state (*collèges privés sous contrat*).² These schools receive substantial public funding: the state pays for the teachers' training and salaries. The school's operating costs are covered by public funds from the national and *département* (sub-regional) levels. Families only pay for the investment costs (renovation or construction of buildings). In contrast to the situation in other education systems, private schools must follow the same curriculum as public schools. In France, private schools are therefore highly dependent on the state. They can be classified as government-dependent private schools, similarly to charter schools in the USA or free schools in England. French private schools are therefore different from fee-funded government-independent schools, such as American private schools or English 'public schools'. In addition to the fact that there has been relatively little research in this area in France, conducting studies on the relative effectiveness of private schools in France on academic achievement is all the more important given that this type of private school has specific characteristics that are distinct even from those of government-dependent private schools in other countries. For example, in charter schools or free schools, teachers do not necessarily have to be certified, whereas in France they are subject to qualification requirements identical to those of the public school system. Moreover, free schools and charter schools cannot select their students, whereas private schools in France can. Finally, in charter schools, the curriculum is flexible (although the schools are held accountable to a performance contract) and free schools do not have to follow the national curriculum, whereas private schools in France must follow the state-defined national curriculum. Because the characteristics of private schools in France differ from those of private schools in other countries, it is important to specifically evaluate their effectiveness in terms of student achievement.

The first authors to focus on the advantage of private schooling on academic success in France were Langouët and Léger (1991). Using a 1973 panel of secondary school students,³ they concluded that private schooling has a positive advantage, especially for children from disadvantaged social backgrounds. Studies using the 1989 panel of secondary school students have yielded more nuanced results. For example, Tavan (2004) shows that private and public students in the panel had the same probability of reaching the eighth grade, but that private school students were more likely to repeat a grade than public school students. Conversely, Caille (2004) shows that

enrollment in private schools led to a higher level of qualifications and a greater chance of reaching 12th grade and obtaining the *baccalauréat*.⁴ In a study based on the 1995 panel of secondary school students, Bertola (2017) finds results that are in line with those of Caille (2004). He shows, first, a positive effect of enrollment in a private secondary school on access to higher education and, second, that students enrolled in private schools had higher academic achievement (except students with the highest initial level of ability). More recently, using the 2007 panel of secondary school students and an original measure of academic performance calculated on the difference between the sixth and ninth-grade national exams, Cayouette-Remblière and Moulin (2019) show that students enrolled in private schools gained in performance advantage over this time in comparison to public school students. Other works using primary school panel data have looked at the advantage of private schooling on success in primary education. Valdenaire (2011) finds that private schooling has a disadvantage on sixth-grade test scores and that private schools increase students' likelihood of repeating a grade. Conversely, Tavan (2004) finds an advantage of private primary schooling. She finds that children who completed all of their primary schooling in the private sector perform as well as public pupils on sixth-grade tests scores, but with less grade retention. More recently, Fougère et al. (2017) show that the third grade test scores of students in private and public schools do not significantly differ. With the exception of the last of the cited studies, which use the distance between the parental home and the nearest private and public primary schools as an instrument, all of the other studies carried out in France rely on weak identification strategies. For this reason, these studies cannot be relied upon to accurately capture the effect of private schooling on student achievement.

Since differences in the organization and governance of educational systems lead to differences in student achievement across countries (Woessmann 2016), it is appropriate to extend the analysis to the case of France. In this article, I use a large French database to estimate the effect of enrollment in a private school on academic achievement as measured by test scores in ninth grade. This paper is the first to provide an estimate of the effect of attending a private school on standardized test scores at the end of lower secondary education (*collège*). Since I do not have a natural method of randomly allocating students to private and public schools, or a lottery as in other countries, I base my estimation strategy on PSM. Regardless of the school subject, I find a positive and significant effect of attending a private school on student achievement. These effects vary between 0.193 and 0.222 standard deviations (SDs) higher on standardized tests in ninth grade for boys and between 0.138 and 0.198 SD higher for girls. Turning then to heterogeneity in academic achievement prior to secondary schooling, I show that private school effects are stronger for boys students with a lower initial level of achievement in mathematics, highlighting the private sector's greater effectiveness for the most academically fragile students. A variety of robustness checks confirm the robustness of the results.

The paper is organized as follows. Section 2 presents the context of private schools in France, the data and summary statistics. Section 3 sets out the empirical strategy. Section 4 presents the results, and Section 5 presents the discussion and conclusion.

2. Institutional background, data, and descriptive statistics

2.1. Private schools in France

In France, the school system is divided into four major levels: three years of 'kindergarten', called *maternelle*, five years of 'primary school' or *primaire* (1st to 5th grade), four years of lower secondary education or *collège* (6th to 9th grade), and finally three years of upper secondary education or *lycée* (10–12th grade).

In 2019, 21.5% of lower secondary students in France were enrolled in private schools (author's calculations from MENJ - DEPP 2020). Among these, 98% were enrolled in a private school under contract with the state (*écoles privées sous contrat*) (MENJ - DEPP 2020). The remaining students were

enrolled in independent private schools (*écoles privées hors contrat*). The majority of private schools in France are thus under contract with the state, a status established by the Debré Act of 1959. Since 2006, a portion of public schools, covering 21.4% of lower secondary school students in the public sector in 2019 (author's calculations from MENJ - DEPP 2020), have been designated as priority education schools.⁵

Schooling in a public school is based on residence assignment. The rules for admission to private schools may vary from school to school and are not always explicitly stated (Fack and Grenet 2010). As specified by Boutchénik, Givord, and Monso (2021), the most common criteria used in private schools are the student's academic level and their motivation to enter the school.⁶ The decision on whether or not to admit or not a student rests with the head of the school. Unlike government-dependent private schools in other countries, students are selected to enter private schools, but they do not take admissions tests to qualify for enrollment.

Competition in France between the private and public sectors is distinctive with respect to the international situation (Valdenaire 2011). Private schools under contract receive government funding and are strictly regulated by the state. These schools must follow the same curricula, pursue the same national pedagogical objectives, and apply the same academic standards. Teacher recruitment is subject to the same qualifications, and teachers' salaries are fully funded by the state. There are very few objective differences between public and private schools under contract. The first concerns teacher recruitment. Teachers in both public and private schools under contract are recruited through a national competitive examination. While teachers in public schools are civil servants and are assigned to a school through a strict administrative procedure,⁷ recruitment to teach in private schools under contract is left to the discretion of the school. This freedom in the recruitment of teachers may play a role, particularly in terms of the cohesion of the teaching team. The second noteworthy difference between public schools and private schools under contract concerns the assignment of students. In public schools, students are assigned to a school on the basis of their area of residence (unless they are granted an exemption). Conversely, private schools under contract recruit their students themselves according to their own criteria.

Private school fees in France are relatively low in comparison to other countries and are therefore not a major barrier to student enrollment. Private schools under contract with the state charge their students an average of 500 euros per year in tuition fees (Davezies and Garrouste 2020). For this reason, publicly funded private schools offer a relatively inexpensive option to evade the constraints of strict school zoning (Fack and Grenet 2010).⁸ Although 97% of students enrolled in private lower secondary schools under contract are enrolled in Catholic schools (author's calculations from MENJ - DEPP 2021; SGEN 2021), religious studies have now almost disappeared from the French national curriculum (Valdenaire 2011). Today, the choice of private education is driven by educational rather than religious reasons (Fack and Grenet 2010; Langouët and Léger 1997).

2.2. Data

In this paper, I take advantage of the fact that the 2007 panel of secondary school students is the first to provide both students' scores on the sixth-grade standardized examinations and the scores from the *brevet des collèges*⁹ examination at the end of the ninth grade. The standardized tests upon entry into sixth grade¹⁰ take place on the first days of the new school year and focus on mathematics and French language skills acquired in primary school. The ninth grade *brevet* examination assesses knowledge and skills acquired up to the end of lower secondary education in three school subjects: mathematics, French, and history–geography (including civics). With the 2007 panel data, it is now possible for the first time to measure the effect of private schooling on student achievement at the end of lower secondary school while controlling for students' level of academic achievement at the beginning.¹¹

The 2007 panel of secondary school students is a random sample of 1/22 of all students ($n = 34,986$) who entered sixth grade in 2007 and were followed until 2013. The panel is

constructed by drawing a balanced random sample from a comprehensive sampling frame using available information on students and colleges within the scope of the study in such a way that the sample faithfully reproduces these observed characteristics. The panel provides detailed information on each student's sociodemographic characteristics and school career. In addition to data collected from schools each year, the database includes two surveys of the students' families carried out in 2008 and 2011 and allowing to collect more detailed information on the socio-economic status of the student.¹²

In order to study the effect of attending a private school on student achievement, I only consider students for whom a score is given for the sixth-grade standardized examination¹³ ($n = 27,186$) and who then took the brevet examinations in an academic track¹⁴ four or five years after entering sixth grade¹⁵ with reported scores in mathematics, French, and history–geography ($n = 22,407$). I also exclude students who did not participate in the family surveys in 2007 and 2011 ($n = 20,352$).¹⁶ Moreover, I only consider students who were enrolled exclusively in either public or private school between the sixth and ninth grades. I chose to exclude switchers as previous studies have highlighted the fact that they differ from the population of students who attend only one or the other type of school (Caille 2004; Spees and Lauen 2019). While students attending priority education schools are on average lower in terms of test scores than those in non-priority education schools (Davezies and Garrouste 2020; Cayouette-Remblière and Moulin 2019), I chose to keep them in this study in order to measure the effect of private schooling on educational achievement compared to schooling in the public sector as a whole.¹⁷ My final sample comprises 19,404 students.

2.3. Descriptive statistics

Table 1 presents selected descriptive statistics separately for boys and girls, and for the treatment group (students enrolled in private schools) and the control group (students enrolled in public schools). The extended table of descriptive statistics, including all variables used in the PSM estimation, can be found in Table A.1.

The sociodemographic characteristics used include a set of variables concerning the students' parents: their socio-occupational category, educational attainment, and monthly family income. Fathers in managerial, professional, and higher-level intellectual occupations and those who are manual workers respectively represent 31.55% (30.49%) and 16.23% (19.14%) of boys (girls) in the treatment group versus 17.41% (16.98%) and 33.22% (33.79%) in the control group. There are also marked differences between the groups in the mother's educational attainment. While the mothers of 47.66% (45.32%) of boys (girls) in private schools have a tertiary degree, in public schools the proportion is only 29.12% (26.47%). Moreover, I have chosen to include the family's financial resources as a discrete variable in the models, including a non-response category. There are also considerable differences in monthly family income between students enrolled in private and public schools. While 40.75% (37.88%) of boys (girls) in the treatment group had a monthly family income of more than 4000 euros, this was true of only 21.83% (20.35%) of boys (girls) in the control group.

Boudon's (1974) work on the notion of aspiration¹⁸ in educational choices led me to include a variable intended to capture it. During the sixth-grade year, parents are asked about the desired school orientation for their child: to prepare a general *baccalauréat*, a technical or vocational *baccalauréat*, a vocational certificate, enter the job market or apprenticeship, do not know. Descriptive statistics highlight a significant difference in family aspirations between students in private and public schools. While 54.87% (59.97%) of families in the treatment group want their boy (girl) to obtain a general (academic) *baccalauréat*, the proportion is lower for families in the control group (43.64% for boys and 49.22% for girls).

With regard to the characteristics of the schools, I am interested in the size of the urban area where the school is located. There are significant disparities in the distribution of private schools between urban areas of different sizes, which influences students' probability of going to a

Table 1. Selected descriptive statistics.

	Boys			Girls		
	Treatment group	Control group	<i>p</i> -Value	Treatment group	Control group	<i>p</i> -Value
<i>Sociodemographic characteristics</i>						
Father's socio-occupational category						
Farmer	3.91%	2.07%	0.000	3.01%	2.27%	0.064
Self-employed	16.71%	10.20%	0.000	16.75%	10.07%	0.000
Managers, professionals, and higher-level intellectual Occupations	31.55%	17.41%	0.000	30.49%	16.98%	0.000
Intermediate occupations	18.39%	18.88%	0.645	17.26%	17.78%	0.604
Clerical and sales workers	8.47%	10.91%	0.003	7.55%	11.13%	0.000
Manual workers	16.23%	33.22%	0.000	19.14%	33.79%	0.000
Retired, economically inactive, or NR	4.75%	7.30%	0.000	5.79%	7.98%	0.002
Mother's educational attainment						
Tertiary education	47.66%	29.12%	0.000	45.32%	26.47%	0.000
General upper secondary education	6.43%	6.68%	0.710	7.72%	6.59%	0.086
Technical or vocational upper secondary education	13.52%	11.61%	0.029	11.64%	11.47%	0.840
Vocational education	18.93%	26.17%	0.000	21.75%	26.63%	0.000
No qualification or NR	13.46%	26.42%	0.000	13.57%	28.84%	0.000
Monthly family income						
<1000 euros	1.02%	4.51%	0.000	1.93%	4.76%	0.000
[1000; 1200] euros	1.86%	3.78%	0.000	1.99%	4.49%	0.000
[1200; 1500] euros	3.43%	7.13%	0.000	3.46%	7.32%	0.000
[1500; 1800] euros	3.67%	6.90%	0.000	4.32%	7.44%	0.000
[1800; 2000] euros	2.64%	4.80%	0.000	2.73%	4.95%	0.000
[2000; 2500] euros	7.15%	12.48%	0.000	9.14%	12.71%	0.000
[2500; 3000] euros	10.46%	13.56%	0.001	11.98%	13.12%	0.196
[3000; 4000] euros	23.02%	20.20%	0.010	20.95%	19.35%	0.123
[4000; 6000] euros	23.68%	15.90%	0.000	22.09%	14.91%	0.000
>6000 euros	17.07%	5.93%	0.000	15.79%	5.44%	0.000
NR	6.01%	4.82%	0.043	5.62%	5.51%	0.855
<i>Family's attitudes toward school</i>						
Desired school orientation						
Prepare a general <i>baccalauréat</i>	54.87%	43.64%	0.000	59.97%	49.22%	0.000
Prepare a technical or vocational <i>baccalauréat</i>	6.55%	9.68%	0.000	4.43%	6.75%	0.000
Prepare a vocational certificate	1.74%	4.53%	0.000	1.08%	4.24%	0.000
Enter the job market or apprenticeship	0.84%	2.12%	0.001	0.28%	1.44%	0.000
Do not know or NR	36.00%	40.02%	0.002	34.24%	38.35%	0.001
<i>School characteristics</i>						
Size of the urban area						
Rural	4.99%	10.54%	0.000	5.74%	10.51%	0.000
<5000 inh	8.11%	11.92%	0.000	8.57%	12.03%	0.000

(Continued)

Table 1. Continued.

	Boys			Girls		
	Treatment group	Control group	<i>p</i> -Value	Treatment group	Control group	<i>p</i> -Value
[5000; 10,000] inh	8.17%	9.81%	0.039	8.18%	10.07%	0.015
[10,000; 20,000] inh	9.38%	8.22%	0.124	8.69%	8.23%	0.523
[20,000; 50,000] inh	10.04%	8.60%	0.061	9.54%	8.81%	0.327
[50,000; 100,000] inh	9.86%	7.33%	0.000	8.86%	7.96%	0.209
[100,000; 200,000] inh	7.45%	6.73%	0.292	7.50%	7.12%	0.582
[200,000; 2,000,000] inh	26.44%	21.58%	0.000	26.46%	20.94%	0.000
Paris agglomeration	15.56%	15.26%	0.757	16.47%	14.33%	0.022
<i>Educational characteristics of the student</i>						
Enrolled in a private primary school	65.75%	5.20%	0.000	64.96%	5.32%	0.000
Grade retention in primary school	5.53%	9.70%	0.000	4.94%	8.95%	0.000
Sixth-grade test score in mathematics (100)	74.58	69.28	0.000	69.55	64.51	0.000
Sixth-grade test score in French (100)	60.64	54.49	0.000	66.02	60.76	0.000
Number of observations	1664	7724	1761	8255		

Notes: The values reported are the sample means (except for test scores); *p*-values are for *t*-tests of equal means. Additional variables in Table A.1: mother's socio-occupational category, father's educational attainment, cultural capital, family type, parents' countries of birth, number of siblings. Source: 2007 panel of secondary school students in France, MEN-DEPP.

private school. For example,¹⁹ 24.60% of schools in urban areas with between 200,000 and 2,000,000 inhabitants are private schools, versus only 13.66% in rural areas (author's calculations from the 2007 schools database). In the sample, 4.99% (5.74%) of boys (girls) in the treatment group were in a school located in an urban area versus 10.54% (10.51%) of boys (girls) in the control group.

Regarding the student's educational characteristics, the analysis includes standardized test scores in sixth grade and enrollment in a private primary school for at least one year. The descriptive statistics highlight this fact: 65.75% (64.96%) of boys (girls) who were enrolled in a private secondary school had previously been enrolled in a private primary school²⁰ versus only 5.20% (5.32%) of students enrolled in a public secondary school. Moreover, as in the existing literature (see, for example, Booker et al. 2011; Vandenberghe and Robin 2004; Witte 1992), I use standardized test scores in sixth grade to control for students' initial level of academic achievement. In mathematics (French), the average scores (out of 100) of boys and girls in private schools were 74.55 (60.64) and 69.55 (66.02), respectively, while for public school students they were 69.28 (54.49) for boys and 64.51 (60.76) for girls.

Figure 1 presents the distribution of the three outcomes (ninth-grade test scores in mathematics, French, and history–geography) separately for boys and girls in private and public schools. Boys (girls) attending a private school scored about 0.467 (0.404) SD higher on average in mathematics, 0.392 (0.368) SD higher in French, and 0.382 (0.319) SD higher in history–geography compared to students in public school. The large differences in scores between public and private school students could be due to disparities in sociodemographic characteristics, family's attitudes toward school, school characteristics, or the educational characteristics of the students. In the next section, I present the methodology used to measure the effect of private schooling by matching students in public and private schools with an equal probability of enrolling in private school based on their individual characteristics.

3. Empirical approach

3.1. Estimation strategy

3.1.1. Identification of causal treatment effect

The objective of this paper is to estimate the causal impact of private school attendance on student achievement at the end of lower secondary school in France. To do this, I use PSM.²¹ The analysis used to estimate causal effects is thus based on the framework of Roy (1951) and Rubin (1974). The two potential outcomes are $Y(1)$ for students who attend private school ($D = 1$) and $Y(0)$ for students who attend public school ($D = 0$). The parameter of interest is the average treatment effect on the treated (ATT):

$$\tau_{ATT} = E[Y(1) | D = 1] - E[Y(0) | D = 1].$$

The last term on the right-hand side of this equation gives the hypothetical outcome without treatment for those who received the treatment. As this term is not observed, the fundamental evaluation problem applies. Attempting to avoid this problem using data from non-participants will lead to selection bias, because membership in the treatment and control groups is not randomly assigned – instead, assignment to the treatment and control groups results from differences in (un)observed characteristics. PSM methods correct for this selection bias by balancing observable characteristics across the treatment and control groups. This technique relies on the identifying conditional independence assumption (CIA), which implies that while the unobserved potential outcome may be conditional on observed characteristics, it is independent of treatment assignment (Rosenbaum and Rubin 1983):

$$Y(0) \perp\!\!\!\perp D | P(X),$$

where $P(X) = \Pr(D = 1 | X)$ is the propensity score, i.e. the probability of receiving the treatment

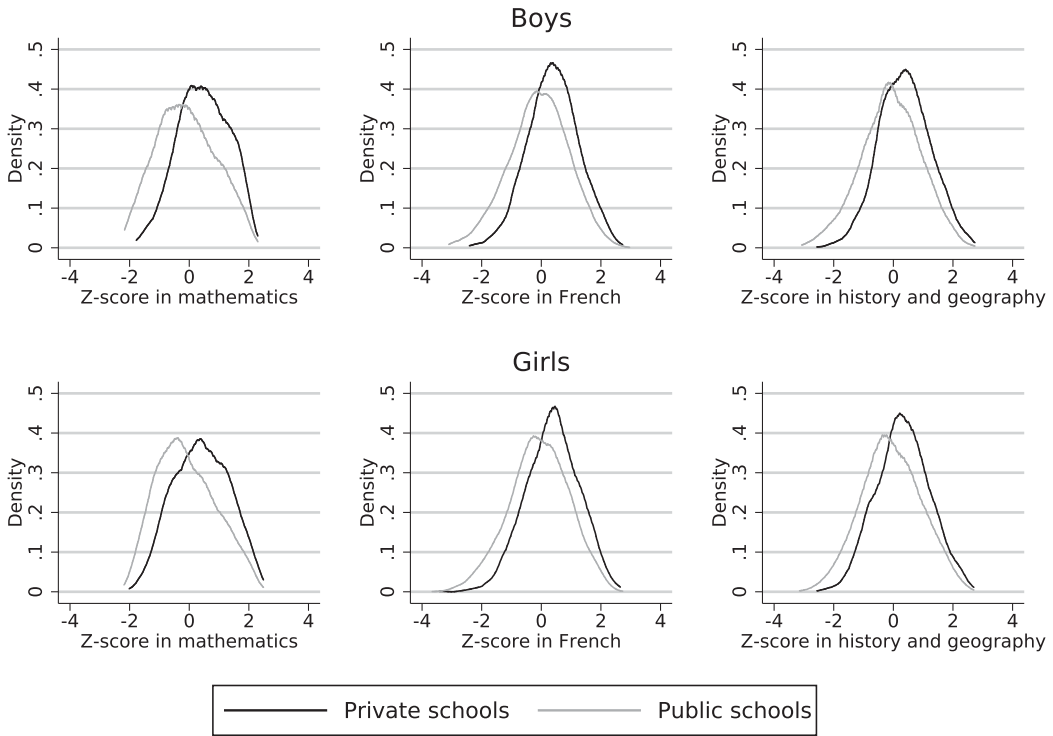


Figure 1. Ninth-grade test scores in mathematics, French, and history–geography. Note: This figure displays the ninth-grade test scores in mathematics, French, and history–geography for the treatment and control group.

Source: 2007 panel of secondary school students in France, MEN-DEPP.

given the observed characteristics X . The CIA is a strong assumption, which implies that the propensity score specification must include all characteristics that simultaneously influence the probability of treatment assignment and the outcome. Propensity score estimation is based on the assumption that selection into the treatment group is based on observable factors. In this study, the variables included in the specification as related to both the choice of private school enrollment and ninth-grade test scores are derived from economic theory, empirical results from other studies, and institutional systems in education in France. The database can be used to control for a large set of variables including sociodemographic characteristics, family's attitudes toward school, school characteristics, and the student's educational characteristics. I consider that, adjusting for these variables, the CIA holds. A further requirement is that the variables included in the specification must not themselves be influenced by the treatment.

In addition, I also make the overlap (or common support) assumption, namely:

$$0 < \Pr(D = 1 | X) < 1, \quad \forall X.$$

This assumption holds that for each value of any given observable characteristic X , everyone in the population of interest has a positive probability both of being treated and not treated.

Assuming that the conditional independence and common support assumptions hold, the PSM estimator for the ATT can be written as

$$\tau_{\text{ATT}} = E[Y(1) | P(X), D = 1] - E[Y(0) | P(X), D = 0].$$

3.1.2. The risk of hidden bias

If there is an unobserved variable that has an effect on both assignment into private school and educational outcomes, a hidden bias might arise to which matching estimators are not robust (Caliendo and Kopeinig 2008). Supposing that private school choice is determined by observable characteristics X and unobserved components U :

$$\Pr(D = 1 | X, U) = F(\beta X + \gamma U),$$

where $F(\cdot)$ is the logistic distribution, if $\gamma = 0$, there is no hidden bias, whereas if $\gamma \neq 0$, two students with the same observed characteristics have different probabilities of being exposed to the treatment. Because the magnitude of this potential bias cannot be estimated with non-experimental data, I assess the sensitivity of my results using a bounding approach (Rosenbaum 2002). Formally, there is a hidden bias if two students i and j with the same observed covariates have different probabilities P of being enrolled in private school. The bounds of the odds ratio that the two matched students will receive treatment can be written as

$$\frac{1}{e^\gamma} \leq \frac{P_i(1 - P_j)}{P_j(1 - P_i)} \leq e^\gamma,$$

where $\Gamma = e^\gamma$ is a measure of the degree of departure from the unconfoundedness assumption. Two matched students i and j have the same probability of being enrolled in a private school only if $e^\gamma = 1$. Otherwise, if $e^\gamma = 2$ for example, two students who appear to be similar in terms of observed characteristics could differ in their odds of receiving the treatment by as much as a factor of 2.

3.2. Implementation of the matching procedure

3.2.1. Propensity score estimation

The first step in the PSM procedure is to estimate the propensity scores using a logit model. The probability of being enrolled in a private school depends not only on the propensity of parents to want to enroll their children in a private school but also on the likelihood that these children will be selected to enter a private school.

First, as explained in Section 2.1, selection into private schools is not random. Controlling for the students' initial level of achievement and grade retention in primary school in the propensity scores allows me to take into account the students' academic level, which is one of the selection criteria for admittance to private schools. The other criteria used by private schools, the student's motivation to enter the school, is usually measured through a cover letter and/or an interview. As explained by Blanchard and Cayouette-Remblière (2016), the most privileged families' social dispositions (ways of being, thinking, communicating, and acting) tend to be closer to those of the school. This allows these families to better understand and respond to the requirements and expectations of educational institutions. This allows them to better prepare their children for an interview, help them write a more effective letter of motivation, etc. By taking into account the students' social origin (parents' socio-occupational category, parents' educational attainment, family income, and cultural capital), I indirectly control for this phenomenon. In addition, family's attitudes towards school and the fact of having been enrolled in a private elementary school also indirectly measure parents' motivation to have their child attend a private school.

Second, parents' propensity to want to enroll their children in a private school is measured by their social origin (Héran 1996), by their attitudes toward school, by having previously enrolled their child in a private school (Bertola 2017), and by the presence of private schools close to their homes (Fougère et al. 2017) (which I measure indirectly through the size of the urban area; see Section 2.3).

Moreover, the model specification includes other variables that are likely to influence students' educational trajectories: parents' social origin and educational attainment (Cayouette-Remblière

and Moulin 2019), parents' income (Evans and Schwab 1995), size of the urban area (Cayouette-Remblière and Moulin 2019), family cultural capital (Bourdieu and Passeron 1964), family structure (Hofferth 2006), number of siblings (Blake 1989), parents' countries of birth (Ichou 2013), and grade retention in primary school (d'Haultfoeuille 2010).

I perform the matching procedure separately for boys and girls.²² The logit estimates of the propensity scores are presented in Table A.2. Certain variables play a determinant role in the probability of attending a private school: family financial resources, the size of the urban area of the school, and previous enrollment in a private primary school. This is in line with the results of previous studies.

Figure 2 plots the distributions of estimated propensity scores for the treated and control groups. It can be seen, first, that the distribution of propensity scores differs greatly between the treatment and control groups prior to matching. This indicates that, before matching, individuals in the treatment group have on average a higher probability of enrolling in private school. The bulge in the propensity score distribution is linked to the difference in the probability of attending a private lower secondary school conditional on enrollment in a private primary school. Second, these distributions show overlap between the two propensity score distributions, with students from each group along the entire distribution of propensity scores, with more limited representation of the treated group at the very beginning of the distributions. I impose a common support condition in order to ensure that I am comparing students with identical propensity scores. This led to the exclusion of 8 boys and 15 girls, representing an entirely acceptable 0.1% of the sample (Caliendo, Hujer, and Thomsen 2015).

Once the propensity scores have been estimated and used to match students in the treatment and control groups, I can estimate the average treatment effects on the treated. To do so, I use Epanechnikov kernel matching with a bandwidth of 0.06. Furthermore, in Section 4.3, I use alternative matching algorithms (nearest neighbor, radius matching, inverse probability weighting (IPW)) and different bandwidths for the kernel estimator. The results are relatively

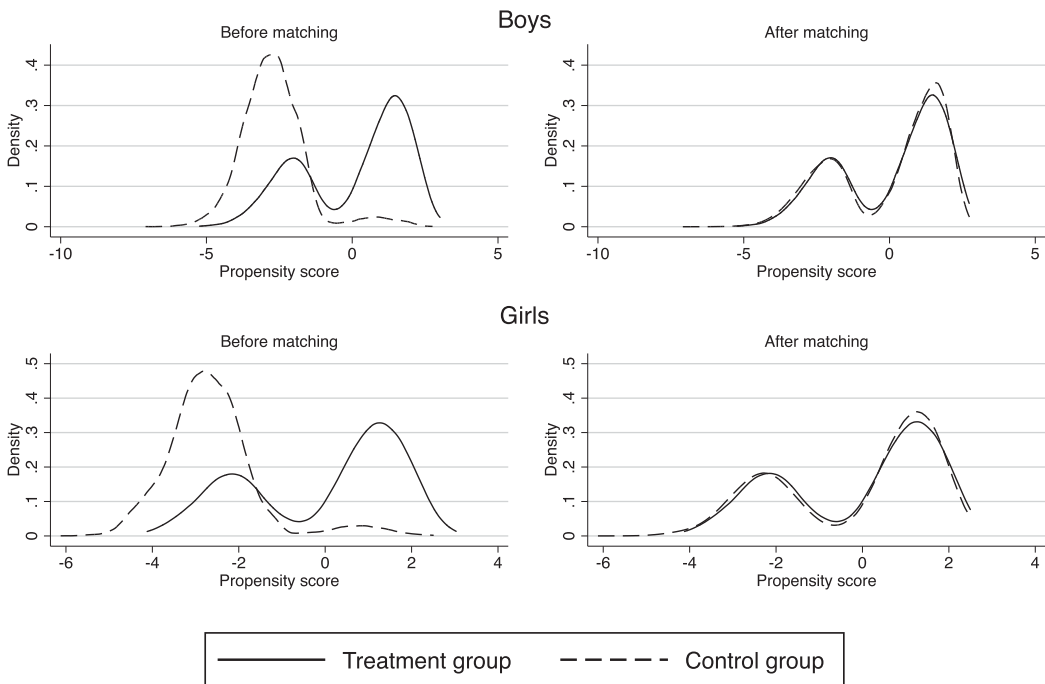


Figure 2. Density balancing plot. Note: This figure displays the propensity score distributions for the treatment and control group before (left side) and after (right side) the matching procedure.

Source: 2007 panel of secondary school students in France, MEN-DEPP.

similar for all types of estimators. Following Caliendo, Mahlstedt, and Mitnik (2017) and Caliendo and Tübbicke (2020), I use bootstrapping with 999 replications to obtain the p -values.

Finally, there is no clear consensus regarding the use of sample weights in PSM. As in other works in the literature (cf., for example, Chudgar and Quin 2012, who makes an identical choice in a similar framework), I choose not to use sample weights.

3.2.2. Matching quality

Once the matching has been carried out, its quality must be validated. Table 2 presents different covariate balance indicators before and after matching. More specifically, I am interested in two indicators. The first is the number of statistically significant differences in means, which, at the 1% significance level, decreased from 43 (39) variables before matching to 0 (0) after matching for boys (girls). The second is the mean absolute standardized bias, which decreased from 14.94% (13.55%) to 2.03% (2.08%) for boys (girls) – below the threshold of 3–5% defined by Caliendo and Kopeinig (2008), which confirms the high quality of the matching procedure. Moreover, the pseudo- R^2 values for propensity score estimates decrease to low levels after matching, and the p -value of joint significance tests goes from extremely significant before matching and to virtually 1 afterward. In summary, the matching procedure appears to be successful in eliminating the difference in the values of the observed variables between the treatment and control groups.

4. Estimation results

4.1. Main results

Table 3 presents the estimated average treatment effects. These results highlight a strong effect of attending a private school on educational achievement. Private school attendance had a large effect on students' ninth-grade test scores in all three school subjects, both for boys and girls.

Table 2. Matching quality indicators.

	Boys		Girls	
	Before matching	After matching	Before matching	After matching
Number of variables with significant differences in means				
At 1% level	43	0	39	0
At 5% level	48	1	43	1
At 10% level	51	2	47	4
Number of variables with absolute standardized bias				
<1%	3	19	4	23
1% to <3%	4	27	7	22
3% to <5%	6	15	10	14
5% to <10%	13	3	8	5
10% to <15%	13	0	18	0
>15%	25	0	17	0
Mean absolute standardized bias in %	14.94	2.03	13.55	2.08
Median absolute standardized bias in %	11.46	1.89	11.23	1.63
Pseudo- R^2 for propensity score estimation	0.389	0.010	0.374	0.009
p -Value of joint significance test	0.000	0.941	0.000	0.979
Total number of variables	64	64	64	64
Treated students off support		8		15

Notes: The following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Source: 2007 panel of secondary school students in France, MEN-DEPP.

In mathematics, boys (girls) who attended a private school scored an average 0.222 (0.198) SD higher than students who attended a public school. In French, the advantage of private school is still sizeable, although smaller than in mathematics: 0.203 SD higher for boys and 0.138 SD higher for girls. In history–geography, boys (girls) enrolled in private school scored 0.193 (0.154) SD more on average compared to those enrolled in public school.²³

These results suggest a substantial advantage in educational achievement for students enrolled in private schools as opposed to public schools. These effects are consistent with those identified in the academic literature (see, for example, Abdulkadiroğlu et al. 2011; Angrist et al. 2002; Angrist, Pathak, and Walters 2013; Chabrier, Cohodes, and Oreopoulos 2016; Dobbie and Fryer 2011, 2015).

4.2. Heterogeneity effects

I continue the analysis by looking at heterogeneity in the effect of enrollment in a private school with students' initial level of academic achievement. In particular, it will help determine which groups of students benefit the most from private schooling in France. The analysis divides students into quartiles by sixth-grade test scores in mathematics and French (history–geography is not tested until the ninth grade). The results are presented in Table 4.

The results show that boys with low initial academic performance (quartile 1) in mathematics make very significant gains in comparison to students with high initial academic performance (quartile 4) on ninth-grade test scores in mathematics or history and geography by attending a private school. Students with a sixth-grade mathematics test score in the lowest quartile scored 0.408 SD higher in mathematics on the ninth-grade test compared to public school students from the same quartile, whereas the private school advantage in the highest quartile score averages only 0.067 SD higher. This is also true, to a lesser extent, for mathematics and history and geography ninth grade test scores, with respect to initial academic performance in French. For girls, this effect is not verified. These effects are consistent with the work of Chabrier, Cohodes, and Oreopoulos (2016), who find a larger effect of private school for previously low-performing students.

4.3. Robustness analysis

After having identified a strong positive effect of private schooling on students' test results in mathematics, French, and history–geography, I need to check the robustness of the main results. A rigorous evaluation is needed to assess the validity of the identifying assumption. Moreover, because the

Table 3. Matching estimation results.

	Mathematics	French	History and geography
<i>Boys</i>			
ATT	0.222***	0.203***	0.193***
Standard error	(0.047)	(0.047)	(0.047)
Treated students off support	8	8	8
Number of observations	9388	9388	9388
<i>Girls</i>			
ATT	0.198***	0.138***	0.154***
Standard error	(0.044)	(0.044)	(0.044)
Treated students off support	15	15	15
Number of observations	10,016	10,016	10,016

Notes: The following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Average treatment effects on the treated (ATT) are calculated using Epanechnikov kernel PSM with bandwidth 0.06. Standard errors are in parentheses. *p*-Values and standard errors are bootstrapped and based on 999 replications. **p* ≤ 0.1. ***p* ≤ 0.05. ****p* ≤ 0.01. Source: 2007 panel of secondary school students in France, MEN-DEPP.

Table 4. Heterogeneity effects.

	Mathematics	French	History and geography
<i>Sixth-grade test score in mathematics</i>			
Boys			
Quartile 1	0.408*** (0.084)	0.253*** (0.100)	0.325*** (0.106)
Quartile 2	0.309*** (0.069)	0.167*** (0.075)	0.232*** (0.077)
Quartile 3	0.197*** (0.082)	0.207*** (0.086)	0.245*** (0.090)
Quartile 4	0.067*** (0.082)	0.119*** (0.082)	0.029 (0.092)
Significant difference between Q1 and Q4	Yes	No	Yes
Girls			
Quartile 1	0.139*** (0.073)	0.105*** (0.095)	0.180*** (0.096)
Quartile 2	0.171*** (0.072)	0.094*** (0.080)	0.233*** (0.081)
Quartile 3	0.176*** (0.090)	0.211*** (0.087)	0.078*** (0.097)
Quartile 4	0.115*** (0.069)	0.004 (0.069)	0.031 (0.078)
Significant difference between Q1 and Q4	No	No	No
<i>Sixth-grade test score in French</i>			
Boys			
Quartile 1	0.279*** (0.099)	0.305*** (0.105)	0.264*** (0.110)
Quartile 2	0.257*** (0.107)	0.194*** (0.099)	0.107*** (0.107)
Quartile 3	0.223*** (0.093)	0.086 (0.079)	0.241*** (0.086)
Quartile 4	0.000 (0.090)	0.121*** (0.079)	0.060 (0.089)
Significant difference between Q1 and Q4	Yes	No	Yes
Girls			
Quartile 1	0.175*** (0.093)	0.131 (0.109)	0.176 (0.111)
Quartile 2	0.178*** (0.085)	0.137*** (0.077)	0.139*** (0.083)
Quartile 3	0.255*** (0.071)	0.177*** (0.059)	0.231*** (0.070)
Quartile 4	0.092*** (0.081)	0.054 (0.068)	−0.008 (0.081)
Significant difference between Q1 and Q4	No	No	No

Notes: The following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Average treatment effects on the treated (ATT) are calculated using Epanechnikov kernel PSM with bandwidth 0.06. Standard errors are in parentheses. p -Values and standard errors are bootstrapped and based on 999 replications. * $p \leq 0.1$. ** $p \leq 0.05$. *** $p \leq 0.01$. Source: 2007 panel of secondary school students in France, MEN-DEPP.

estimates may depend on the choices of matching algorithm and variables, I need to check the sensitivity of the estimates to these choices. To do this, I first assess the validity of my results on the matching estimators under other possible specifications. I also perform robustness checks on two separate samples. I then check the robustness of the results by evaluating the effect of deviation from the identifying assumption.

4.3.1. Modeling approach

To assess the validity of the results with respect to other possible model specifications, I replicate the analysis presented in Table 3 with alternative estimation methods. I then change the bandwidth of

the kernel estimator and try alternative estimators (nearest neighbor, IPW, and radius matching). If the results vary across these different specifications, this implies that they are sensitive to the specification of the model and calls their robustness into question. Table 5 presents the results, which are relatively stable across estimators. All highlight significant statistical differences between students in private and public schools in ninth-grade test scores in mathematics, French, and history–geography. In mathematics, I find a private school effect for boys (girls) ranging from 0.189 (0.149) SD with IPW to 0.321 (0.289) for kernel matching with bandwidth 0.2. For French, I find a private school effect for boys (girls) ranging from 0.161 (0.096) with IPW to 0.280 (0.236) SD for kernel matching with bandwidth 0.2. In history–geography, I find a private school effect for boys (girls) ranging from 0.171 (0.112) SD for IPW to 0.272 (0.232) for kernel matching with bandwidth 0.2. My results concerning the greater effectiveness of private schools for academic achievement are thus not sensitive to the particular estimation method chosen.

4.3.2. Alternative samples

In order to validate the results, I also perform robustness checks on two separate samples.

The first sample distinguishes the students in the control group who were enrolled in priority education institutions from those who were not. As stressed by Cayouette-Remblière and Moulin (2019), students attending a priority education school score much lower on standardized ninth-grade tests than students enrolled in other public schools. Figure A.1 presents the distribution of

Table 5. Sensitivity with respect to the chosen matching approach.

	Mathematics	French	History and geography
<i>Boys</i>			
Baseline estimates	0.222*** (0.047)	0.203*** (0.047)	0.193*** (0.047)
Nearest neighbor (1:1) with replacement	0.264*** (0.074)	0.229*** (0.073)	0.229*** (0.078)
Nearest neighbor (1:3) with replacement	0.218*** (0.060)	0.220*** (0.058)	0.202*** (0.061)
Kernel 2 (bw = 0.02)	0.205*** (0.051)	0.182*** (0.051)	0.184*** (0.051)
Kernel 3 (bw = 0.2)	0.321*** (0.046)	0.280*** (0.046)	0.272*** (0.046)
Radius (caliper = 0.1)	0.282*** (0.047)	0.245*** (0.047)	0.239*** (0.047)
IPW	0.189*** (0.052)	0.161*** (0.044)	0.171*** (0.048)
<i>Girls</i>			
Baseline estimates	0.198*** (0.044)	0.138*** (0.044)	0.154*** (0.044)
Nearest neighbor (1:1) with replacement	0.174*** (0.070)	0.096*** (0.064)	0.131*** (0.066)
Nearest neighbor (1:3) with replacement	0.162*** (0.054)	0.098*** (0.051)	0.130*** (0.052)
Kernel 2 (bw = 0.02)	0.180*** (0.046)	0.116*** (0.046)	0.137*** (0.046)
Kernel 3 (bw = 0.2)	0.289*** (0.042)	0.236*** (0.042)	0.232*** (0.041)
Radius (caliper = 0.1)	0.248*** (0.043)	0.196*** (0.043)	0.200*** (0.043)
IPW	0.149*** (0.039)	0.096*** (0.041)	0.112*** (0.044)

Notes: For matching estimators, the following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Standard errors are in parentheses. *p*-Values and standard errors are bootstrapped and based on 999 replications. **p* ≤ 0.1. ***p* ≤ 0.05. ****p* ≤ 0.01. Source: 2007 panel of secondary school students in France, MEN-DEPP.

the three outcomes (ninth-grade test scores in mathematics, French, and history–geography) separately for boys and girls in priority and non-priority education schools before the matching procedure. Boys (girls) attending a priority education school scored -0.552 (-0.534) SD lower in mathematics, -0.398 (-0.446) SD lower in French, and -0.385 (-0.346) SD lower in history–geography compared to students attending a non-priority education school. The difference in educational achievement between students in the two types of public schools is thus even larger than the gap between students in public and private schools (cf. Figure 2). In Table 6, I reproduce the same PSM estimation as presented in Table 3 but on a subsample excluding public school students who attended a priority education school any time between the sixth and ninth grades. The results still show a strong and significant difference between the performance of students enrolled in private and public non-priority education schools. In comparison to students enrolled in public non-priority education schools, boys (girls) enrolled in private schools scored 0.198 (0.189) SD higher on the standardized ninth-grade test in mathematics, 0.192 (0.138) SD higher in French, and 0.163 (0.161) higher in history–geography. These results call for two observations. The first is that my baseline results are still valid if I reduce the sample to public school students not enrolled in a priority education school. The inclusion of students from priority education institutions in the general analysis does not bias the results. The second is that the exclusion of students enrolled in a priority school seems to benefit girls more, as excluding priority education schools decreases the advantage of private over public schools for boys on all three outcomes.

In France, it is possible to circumvent the school district map for assignment to a public school based on residential location in three ways: by negotiating with the local education authority (*Académie*) on the basis of several criteria (such as need-based grants or health), by declaring a false address, or by enrolling to a private school (Courtieux and Maury 2020). The second sample

Table 6. Matching estimation results on alternative samples.

	Mathematics	French	History and geography
<i>Panel A – excluding priority education</i>			
Boys			
ATT	0.198***	0.192***	0.163***
Standard errors	(0.048)	(0.048)	(0.047)
Treated students off support	8	8	8
Number of observations	7797	7797	7797
Girls			
ATT	0.189***	0.138***	0.161***
Standard errors	(0.045)	(0.045)	(0.046)
Treated students off support	15	15	15
Number of observations	8294	8294	8294
<i>Panel B – excluding students enrolled in the local school district</i>			
Boys			
ATT	0.221***	0.141***	0.149***
Standard errors	(0.092)	(0.100)	(0.094)
Treated students off support	41	41	41
Number of observations	2490	2490	2490
Girls			
ATT	0.180***	0.177***	0.122***
Standard errors	(0.081)	(0.085)	(0.087)
Treated students off support	32	32	32
Number of observations	2723	2723	2723

Notes: The following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Average treatment effects on the treated (ATT) are calculated using Epanechnikov kernel PSM with bandwidth 0.06. Standard errors are in parentheses. p -Values and standard errors are bootstrapped and based on 999 replications. * $p \leq 0.1$. ** $p \leq 0.05$. *** $p \leq 0.01$. Source: 2007 panel of secondary school students in France, MEN-DEPP.

distinguishes the students in the control group whose parents adhered to the official map for school enrollment in sixth grade from those who bypassed it. Figure A.2 presents the distribution of the three outcomes (ninth-grade test scores in mathematics, French, and history–geography) separately for boys and girls enrolled in same-district schools and other-district schools before the matching procedure. There are no differences in ninth-grade educational achievement between students in these two groups. Boys (girls) attending a school in their local school district scored -0.085 (-0.070) SD lower in mathematics, -0.075 (-0.069) SD lower in French, and -0.077 (-0.065) SD lower in history–geography compared to students whose parents bypassed the school district. I reproduce PSM estimation for this sample in Table 6. There is still a strong and significant difference between the performance of students enrolled in private and other-district public schools. This means that the effect of private schooling does not result only from parents' ability to choose a school – which also characterizes cases where parents bypass the school districting system – but also from the greater effectiveness of the private education system.

4.3.3. Sensitivity analysis

PSM depends on the validity of the CIA. If there is any unobserved variable that simultaneously affects both the treatment assignment and the outcome, this assumption does not hold, and there is a hidden bias. I use a bounding approach (Rosenbaum 2002), which simulates an unobserved characteristic to test the degree of unobserved heterogeneity to which the results are robust. Table 7 presents the upper bound on the significance level of the effect of being enrolled in a private school in the case of endogenous selection. The robustness of the results to hidden bias varies across boys and girls and the three outcomes. The smallest critical level of hidden bias Γ is 1.35–1.40 for girls in French, and the largest is 1.70–1.75 for boys in mathematics. A critical level of Γ between 1.70 and 1.75 means that an unobserved variable would need to increase the odds of receiving treatment by a factor of 1.75 to undermine the results of matching analysis. For example, for boys (girls) the critical Γ of 1.75(1.55) in mathematics is equivalent to an increase²⁴ in 15.71(12.50) percentage points in private primary school attendance required to make the estimated effect of lower secondary private school attendance insignificant at the 10% level. The results thus seem highly likely to be robust to unobserved heterogeneity.

I also use a simulation approach (Nannicini 2007; Ichino, Mealli, and Nannicini 2008) to better assess the influence of potential unobserved heterogeneity. This method consists of simulating the impact of an unobserved variable U on the outcomes assuming its distribution is similar to that of an observable characteristic X . The results of this simulation-based sensitivity analysis are presented in Table 8. For example, for boys, the simulated confounder that mimics the distribution of grade retention in primary school has a negative effect both on the probability of being enrolled in a secondary private school ($0.713 < 1$) and on ninth-grade test scores in mathematics ($0.122 < 1$). The estimated coefficients for ATTs are slightly larger than in the baseline estimate. These findings suggest that none of the simulated confounders would change the estimated effects of private school attendance on educational achievement. This confirms the robustness of the baseline estimate.

5. Discussion and conclusion

The question of the relative effectiveness of private and public education systems is at the heart of public policy issues in many countries. While the international literature is relatively rich, generally highlighting a positive effect of private school attendance on academic success, empirical work on the subject in the French case is lacking, particularly with regard to lower secondary education. Using data from a national panel of secondary school students, I estimate the effect of enrollment in a private lower secondary school on academic achievement on ninth-grade test scores in three school subjects (mathematics, French, and history–geography). In the absence of a natural randomization of students between private and public schools, I use PSM. The results show a strong and

Table 7. Sensitivity to unobserved heterogeneity – bounding approach.

Γ	Boys			Girls		
	Mathematics	French	History and geography	Mathematics	French	History and geography
1.00	0.000	0.000	0.000	0.000	0.000	0.000
1.25	0.000	0.000	0.000	0.000	0.003	0.001
1.50	0.000	0.001	0.039	0.052	0.685	0.491
1.75	0.191	0.325	0.807	0.859	0.999	0.997
2.00	0.915	0.964	0.999	1.000	1.000	1.000
<i>Critical values</i>						
1%	1.60–1.65	1.55–1.60	1.45–1.50	1.40–1.45	1.25–1.30	1.30–1.35
5%	1.65–1.70	1.60–1.65	1.50–1.55	1.45–1.50	1.30–1.35	1.35–1.40
10%	1.70–1.75	1.65–1.70	1.50–1.55	1.50–1.55	1.35–1.40	1.35–1.40

Notes: The following variables are included in the logit estimation when estimating the propensity score: mother's socio-occupational category, father's socio-occupational category, mother's educational attainment, father's educational attainment, monthly family income, cultural capital, family type, parents' countries of birth, desired school orientation, size of the urban area, enrollment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, and sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table A.1. Reported values correspond to upper bound on the p -value for different values of Γ . Critical values are related to the exact values beginning with which the results become insignificant. Source: 2007 panel of secondary school students in France, MEN-DEPP.

Table 8. Sensitivity to unobserved heterogeneity – simulation approach.

	Mathematics			French			History and geography		
	Outcome effect	Selection effect	ATT	Outcome effect	Selection effect	ATT	Outcome effect	Selection effect	ATT
<i>Boys</i>									
No confounder			0.227			0.202			0.194
Mother's socio-occupational category – clerical and sales workers	0.606	0.695	0.269	0.631	0.676	0.238	0.666	0.662	0.230
Father's socio-occupational category – manual workers	0.456	0.427	0.246	0.405	0.424	0.213	0.526	0.415	0.209
Mother's educational attainment – tertiary education	3.748	1.895	0.244	3.616	1.961	0.214	3.083	2.014	0.203
Father's educational attainment – tertiary education	3.835	2.019	0.244	3.901	2.125	0.213	2.782	2.178	0.204
Monthly family income – >6000 euros	4.382	2.776	0.256	4.219	2.954	0.226	3.197	3.018	0.217
Family type – two parents	1.751	1.703	0.266	1.420	1.767	0.238	1.337	1.780	0.231
Desired school orientation – do not know or NR	0.653	0.905	0.276	0.604	0.904	0.246	0.659	0.899	0.237
Size of the urban area – Paris agglomeration	0.724	1.038	0.278	1.144	1.003	0.247	0.866	1.038	0.239
Enrolled in a private primary school	1.095	36.301	0.294	0.986	37.152	0.277	1.134	36.593	0.237
Grade retention in primary school	0.122	0.713	0.271	0.138	0.662	0.239	0.217	0.642	0.230
<i>Girls</i>									
No confounder			0.194			0.133			0.154
Mother's socio-occupational category – clerical and sales workers	0.594	0.737	0.243	0.666	0.719	0.184	0.612	0.718	0.187
Father's socio-occupational category – manual workers	0.457	0.513	0.230	0.462	0.502	0.169	0.480	0.492	0.170
Mother's educational attainment – tertiary education	4.206	1.975	0.211	3.924	2.037	0.150	3.266	2.113	0.149
Father's educational attainment – tertiary education	4.476	2.020	0.214	3.913	2.090	0.152	3.569	2.142	0.151
Monthly family income – >6000 euros	4.561	2.840	0.223	4.559	2.858	0.162	3.560	2.956	0.166
Family type – two parents	1.882	1.502	0.242	1.444	1.571	0.184	1.504	1.565	0.187

Desired school orientation – do not know or NR	0.685	0.894	0.247	0.612	0.893	0.187	0.641	0.875	0.192
Size of the urban area – Paris agglomeration	0.754	1.217	0.249	1.125	1.154	0.188	0.874	1.175	0.195
Enrolled in a private primary school	1.171	33.665	0.230	1.237	34.665	0.163	0.991	34.952	0.212
Grade retention in primary school	0.116	0.680	0.243	0.154	0.659	0.183	0.164	0.616	0.186

Notes: Average treatment effects on the treated (ATT) are calculated using kernel PSM estimates. The estimation is based on 200 replications. Source: 2007 panel of secondary school students in France, MEN-DEPP.

significant effect of private school attendance on academic success in all three school subjects. Among private school students, boys scored between 0.193 and 0.222 SD higher on standardized tests in ninth grade, and girls between 0.138 and 0.198 SD higher. A set of robustness checks confirm the robustness of the effects. In addition, it is boys students with low academic performance in sixth grade who benefit the most from the positive effects of enrollment in a private lower secondary school.

While these effects are large, they are consistent with the results of other work on the effectiveness of government-dependent private schools in the academic literature. Using student assignment lotteries to estimate the effect of charter school attendance on student achievement in Boston, Abdulkadiroğlu et al. (2011) show that the impact of charter attendance for middle school students is 0.198 SD in English language arts (ELA) and 0.359 SD in mathematics. Also using a lottery-based analysis in Massachusetts, Angrist, Pathak, and Walters (2013) show that urban charter schools improve middle school student achievement by 0.075 SD in ELA and 0.213 SD in mathematics, compared to students in traditional urban public schools. In New York City, Dobbie and Fryer (2013) find an effect of charter schools on mathematics achievement at the end of middle school of about 0.126 SD. In England, looking at the introduction of academy schools, Eyles, Hupkau, and Machin (2016) show a performance advantage of 0.110 SD among students in converted academy schools at the end of Year 11 compared to pupils in schools that did not convert into an academy during the sample period. The effects of attending a private lower secondary school in France that I have highlighted in this article are thus comparable in magnitude to those of government-dependent private schools in other countries.

The similarity of institutional settings of private and public schools in France provides us confidence in the validity of our results. In France, private schools are strictly regulated by the state and receive state funding. Private school teacher recruitment is subject to the same qualification requirements as in the public sector. As in the public sector, teacher salaries in private schools are funded by the state. Moreover, private and public schools apply the same academic standards, and must follow the same curricula and national pedagogical objectives. These results are thus not due to differences in formal academic structures, curricula, and standards, but to the greater effectiveness of the private education system for student achievement. Under these circumstances, how should the better academic performance observed among private school students be explained?

We can immediately exclude a factor whose effectiveness in terms of school success has been the subject of numerous studies in the academic literature, including in France: class size (Angrist and Lavy 1999; Bressoux, Kramarz, and Prost 2009; Gary-Bobo and Mahjoub 2013; Piketty and Valdenaire 2006). Private school classes are on average larger than public school classes, with a mean of 27.1 students in private school classes, compared to 24.6 in public schools (MENJ - DEPP 2020). Moreover, teachers' qualifications, which in theory could help students to attain higher test scores (Lai, Sadoulet, and de Janvry 2011), are lower on average in private schools.²⁵ Similarly, the student-teacher ratio is higher in public schools: 14.4 in private lower secondary schools and 12.9 in public lower secondary schools (MENJ - DEPP 2021). Therefore, other factors must explain the greater academic success of students in private schools in France. The first of these could be discretionary teacher recruitment by the head of the school. This relative freedom in the composition of the teaching team may allow for greater cohesion of the teams around the school's educational project. In private schools, there is a real educational community in which parents genuinely participate, whereas in public schools parents are kept on the sidelines. Another factor that may explain private schools' greater success is their specific social composition. Students whose parents are in managerial, professional and higher-level intellectual occupations, or who have a tertiary degree, are largely overrepresented in private schools (see Table A.1). While the students' sociodemographic characteristics are taken into account in my estimation strategy, it is not possible to control here for such effects of the social composition of classes and schools, which may nonetheless have effects on students' educational success. Identifying precisely the

effects of these different characteristics on the educational success of students in private schools is an area for further research.

While private schools enable their students to achieve better academic results at a relatively low cost, not all students attend private schools. There are several reasons for this phenomenon. First of all, public schools may be located closer to students' homes. Second, at the local level, a public school may have a better reputation, and even better results, than a private school. Third, some parents may not want their children to attend private schools out of personal conviction. Fourth, the number of private schools is limited by the government, and not all families who wish to send their children to a private school can do so: hence selective admittance to these schools. All of these reasons contribute to explaining why, despite the fact that private schools produce better student academic outcomes, there are not currently more students in these schools.

Finally, while private schools do better in terms of effectiveness as measured by test scores, a key question is whether beyond these private benefits, private schools achieve the same level of long-term social benefits as public schools. This question is crucial given that the effects of education are not limited to private returns: the academic literature has highlighted the positive externalities of education (see, for example, Aghion and Howitt 2009; Grossman 2006).

Notes

1. Matching is one of the methods based on selection on observables. Other popular approaches are regression-adjustment, reweighting, and the doubly robust estimator (Cerulli 2015).
2. For ease of reading, in the remainder of the paper, private schools under contract with the state are referred to simply as 'private schools'. Independent private schools are excluded from this study.
3. The panel year refer to the year in which students enter sixth-grade.
4. The *baccalauréat* is the French national exam at the end of secondary school.
5. Priority education in France includes two categories of schools. In 2006, they were classified as RAR (*Réseaux ambition réussite*), the schools with the most socially and academically disadvantaged category of students, and RRS (*Réseaux de réussite scolaire*), for schools with less disadvantaged students (Cayouette-Remblière and Moulin 2019). In 2011, RAR became ECLAIR (*Ecoles, collèges et lycées pour l'ambition, l'innovation et la réussite* (schools for ambition, innovation, and success)); in 2015, they became REP+ (*Réseaux d'éducation prioritaire renforcée* (reinforced priority education networks)), while the RRS schools were renamed REP (*Réseaux d'éducation prioritaire* (priority education networks)).
6. Without mentioning motivation by name, Fack and Grenet (2010) refer to a similar criterion as 'good behavior' and 'commitment to the school's values'.
7. The assignment of a teacher to a public school depends on a score that is calculated on the basis of results in competitive recruitment examinations, seniority, personal situation (disability, request for transfer to reunify with a spouse who works elsewhere), or having taught in a priority education school.
8. Furthermore, this avoidance of the public sector in favor of the private sector is not without consequences for the social composition of schools. Thus, from a study of student enrollment in three French urban areas, Boucht-énik, Givord, and Monso (2021) show that a higher proportion of the most advantaged families avoid public school catchment areas by sending their children to private school, thus contributing to an increase in school segregation between lower secondary schools.
9. The *brevet des collèges* is the French national exam at the end of lower secondary education.
10. These exams do not determine access to grade 6. They are used as an assessment measure to detect students in difficulty.
11. Before the 2007 panel, standardized test scores were not available for either sixth or ninth grade. However, using scores obtained during in-classroom assessment within schools would lead to incorrect results as Cayouette-Remblière and Moulin (2019) showed.
12. The variables for father's and mother's socio-occupational categories, monthly family income, family type, and number of siblings were constructed from the 2011 family survey. Although the survey took place during treatment, this is not a problem for PSM insofar these variables cannot be influenced by the treatment. The variable includes permanent and transitory income; the data set does not allow one to be distinguished from the other. In England, Anders et al. (2020) highlighted a distinct and larger effect of permanent over transitory income on private school participation.
13. This practice is common in the literature, and the inclusion of students with missing baseline scores can change the results (Krueger and Zhu 2004). Among these students, students in private schools are slightly overrepresented (25.9% versus 21.4% for students in public schools).

14. I have chosen to limit my sample to students who are enrolled in an academic track, because including the other tracks in the model would create a specific outcome which would no longer be a variable measured at the national level, as are ninth-grade test scores, but one which would instead be strongly dependent on the local context, since the decisions to orient students into different tracks are defined locally by the teaching teams.
15. I include the scores of students who repeated a grade in lower secondary school in the model, but I do not create a specific variable for grade retention during lower secondary education. This is because estimation based on PSM requires choosing explanatory variables that are not influenced by the treatment, whereas grade retention takes place after the assignment of the treatment. Grade retention policies are decided at a local level in the schools themselves (Blanchard and Cayouette-Remblière 2016), and there are disparities in these policies between public and private schools. For more on grade retention in France, I refer the reader to Gary-Bobo, Goussé, and Robin (2016).
16. Students who did not participate in the family surveys do not show a significant difference in terms of private school attendance (15.5% for students in private schools and 15.7% for students in public schools).
17. There are very few private schools in the priority education program that are included in the analysis.
18. For a recent work on aspirations on educational pathways, I refer the reader to Guyon and Huillery (2021).
19. In addition to the 2007 panel of secondary school students, I use the 2007 version of the national database on schools and students (*Base Centrale Scolarité*) to calculate the proportion of private schools in municipalities of different population sizes (nine categories).
20. However, this does not mean that all students who attend a private elementary school will attend a private lower secondary school. This phenomenon is linked to two elements. The first is the availability of places in lower secondary private schools, and the fact that students, whether or not they were enrolled in an elementary private school, may not be admitted. The second pertains to the proximity of a public secondary school to the parental home and the parents' preference (for reasons of reputation) to send their child there.
21. The implementation of the PSM procedure used here is as specified by Caliendo and Künn (2011), Caliendo, Künn, and Weißenberger (2016), and Caliendo and Tübbicke (2022).
22. This choice is based on the widely discussed gap in educational achievement between boys and girls (Fryer and Levitt 2010; Machin and McNally 2005; Machin and Pekkarinen 2008). Moreover, this practice is commonly used in the literature to analyze differences in the effect of the variable of interest on the outcomes of girls and boys (see Caliendo, Hujer, and Thomsen 2015; Caliendo, Künn, and Weißenberger 2016, for example).
23. This corresponds to a difference in scores between private and public school students ranging from 2.163/100 for girls in French to 4.962/100 for boys in mathematics. Compared to the average score, this difference is quite small for girls in French (average score of 57.257/100), whereas it is relatively large for boys in mathematics (average score of 48.466/100).
24. I compute hidden bias equivalents following DiPrete and Gangl (2004).
25. In France, a distinction is made between three categories of teachers: *agrégé* teachers, who are recruited through a highly selective competitive examination; the *agrégation*, certified teachers who are recruited through a less selective competition; and other teachers who are not recruited through competitive examinations. In theory, having more highly qualified teachers could help improve students' academic performance. And yet in France, 13% of public school teachers are *agrégés* versus only 4% in private schools (author's calculations from MENJ - DEPP 2020).

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